

Student Name:

FRENSHAM

2014

YEAR 12

TRIAL HSC EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time - 5 minutes
- Working time - 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11-14

Total marks - 70

Section I

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II

60 marks

- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Section I**10 marks****Attempt Questions 1 - 10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1-10

1 What is the domain and range of $y = \frac{1}{2} \cos^{-1}\left(\frac{x}{2}\right)$?

(A) Domain: $-2 \leq x \leq 2$ Range: $0 \leq y \leq \pi$

(B) Domain: $-1 \leq x \leq 1$ Range: $0 \leq y \leq \pi$

(C) Domain: $-2 \leq x \leq 2$ Range: $0 \leq y \leq \frac{\pi}{2}$

(D) Domain: $-1 \leq x \leq 1$ Range: $0 \leq y \leq \frac{\pi}{2}$

2 When a polynomial $P(x) = x^3 + ax + 1$ is divided by $(x + 2)$ the remainder is 5. What is the value of a ?

(A) -6

(B) -3.5

(C) 2

(D) 3

3 Which of the following is an expression for $\int \frac{x}{(2-x^2)^3} dx$?

Use the substitution $u = 2 - x^2$.

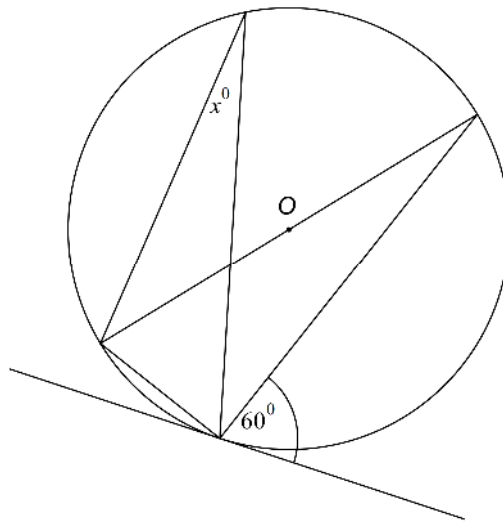
(A) $\frac{1}{2(2-x^2)^2} + C$

(B) $\frac{1}{4(2-x^2)^2} + C$

(C) $\frac{1}{4(2-x^2)^4} + C$

(D) $\frac{1}{8(2-x^2)^4} + C$

- 4 What is the exact value of the definite integral $\int_0^1 \frac{1}{x^2+1} dx$?
- (A) $\frac{\pi}{4}$
- (B) $\frac{\pi}{3}$
- (C) $\frac{\pi}{2}$
- (D) π
- 5 How many arrangements of all of the letters of the word PROBABILITY are possible?
- (A) 362 880
- (B) 9 979 200
- (C) 19 958 400
- (D) 39 916 800
- 6 Find the acute angle between the lines $y = 2x$ and $x + y - 5 = 0$. Answer correct to the nearest degree.
- (A) 18°
- (B) 32°
- (C) 45°
- (D) 72°
- 7 Find the value of x :



- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

- 8 What are the coordinates of the point that divides the interval joining the points $A(-1, 2)$ and $B(3, 5)$ externally in the ratio 3:1?
- (A) $(2.5, 4.25)$
(B) $(2.5, 6.5)$
(C) $(5, 4.25)$
(D) $(5, 6.5)$
- 9 What is the solution to the inequality $\frac{2x-5}{x-4} \geq x$?
- (A) $x \leq -1$ and $4 \leq x \leq 5$
(B) $x \leq -1$ and $4 < x \leq 5$
(C) $x \leq 1$ and $4 \leq x \leq 5$
(D) $x \leq 1$ and $4 < x \leq 5$
- 10 The velocity of a particle moving in a straight line is given by $v = 2x + 5$, where x metres is the distance from fixed point O and v is the velocity in metres per second. What is the acceleration of the particle when it is 1 metre to the right of O ?
- (A) $a = 7 \text{ ms}^{-2}$
(B) $a = 12 \text{ ms}^{-2}$
(C) $a = 14 \text{ ms}^{-2}$
(D) $a = 24 \text{ ms}^{-2}$

Section II

60 marks

Attempt Questions 11 □ 14

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet.

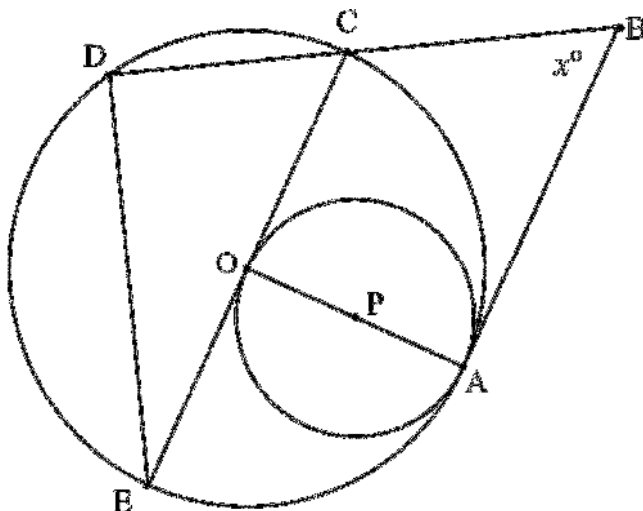
All necessary working should be shown in every question.

Question 11 (15 marks)

Marks

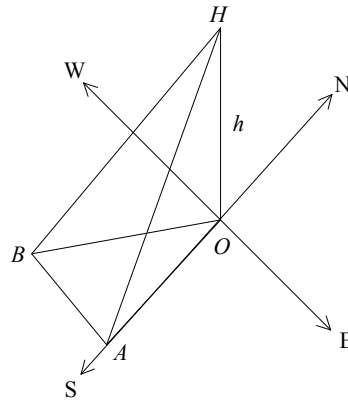
- (a) What are the roots of the equation $4x^3 - 4x^2 - 29x + 15 = 0$ given that one root is the difference between the other two roots? 3
- (b) A circle, centre O , passes through the points A , C , D and E .
 Another circle, centre P , passes through the points A and O .
 CE is a tangent to the circle centre P , with point of contact at O .
 AB is a tangent to both circles with point of contact at A .
 $\angle CBA = x^\circ$.

Show that $\angle CED = (90 - x)^\circ$



- (c) Prove the following identity 2
- $$\frac{\sin \theta}{\sin \theta + \cos \theta} + \frac{\sin \theta}{\cos \theta - \sin \theta} = \tan 2\theta$$
- (d) A class consists of 10 boys and 12 girls. How many ways are there of selecting a committee of 3 boys and 2 girls from this class? 2

- (e) Point A is due south of a hill and the angle of elevation from A to the top of the hill is 35° . Another point B is a bearing 200° from the hill and the angle of elevation from B to the top of the hill is 46° . The distance AB is 220 m.



- | | | |
|------|--|----------|
| (i) | Express OA and OB in terms of h . | 2 |
| (ii) | Calculate the height h of the hill correct to three significant figures. | 2 |
-
- | | | |
|-----|---------------------------------|----------|
| (f) | Factorise $x^3 + 3x^2 - 9x + 5$ | 2 |
|-----|---------------------------------|----------|

Question 12 (15 marks)**Marks**

- (a) The tangent at the point $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ cuts the x -axis at A and the y -axis at B .
- Find the coordinates of M , the midpoint of A and B in terms of P . **2**
 - Show that the locus of M is a parabola. **1**
 - Find the coordinates of the focus of this parabola and the equation of the directrix. **1**
- (b) Use the principle of mathematical induction to prove that for all positive integers n :
- $$1 + 2 + 4 + \dots + 2^{n-1} = 2^n - 1$$
- (c) Find the exact value of $\sin\left[\cos^{-1}\frac{2}{3} + \tan^{-1}\left(-\frac{3}{4}\right)\right]$ **2**
- (d) Find all the angles θ with $0 \leq \theta \leq 2\pi$ for which $\sin\theta + \cos\theta = 1$. **3**
- (e) The function $f(x)$ is given by $f(x) = \sin^{-1}x + \cos^{-1}x$, $0 \leq x \leq 1$.
- Find $f'(x)$. **1**
 - Sketch the graph of $y = f(x)$. **2**

Question 13 (15 marks)**Marks**

- (a) (i) Show that the function $f(x) = xe^x - 1$ has a zero between $x = 0$ and $x = 1$. **1**
- (ii) Using $x = 0.5$ as the first approximation, use Newton's Method to obtain a second approximation. Answer correct to 2 decimal places. **2**
- (b) A golfer hits a golf ball to clear a 6 metres high tree. The tree is 20 metres away on level ground. The golfer uses a golf club that produces an angle of elevation of 40° . Take $g = 10 \text{ ms}^{-1}$.
- (i) Derive the expressions for the vertical and horizontal components of the displacement of the ball from the point of projection. **3**
- (ii) Find the Cartesian equation of the flight path? **2**
- (iii) Calculate the speed at which the ball must leave the ground to just clear the tree. Answer correct to one decimal place. **2**
- (c) Consider the curve $f(x) = (x - 2)^2$
- (i) If the domain is to be restricted to the largest possible domain that contains $x = 0$, so that an inverse function will exist, state the domain. **1**
- (ii) What is the domain of $f^{-1}(x)$? **1**
- (iii) What is the equation of $f^{-1}(x)$? **1**
- (iv) Explain why $x = (x - 2)^2$ gives the points of intersection of $y = f(x)$ and $y = f^{-1}(x)$ and hence why $x = 1$ is the only point of intersection. **2**

Question 14 (15 marks)**Marks**

(a) Find $\int \cos^2 2x dx$

2

(b) A particle moves in a straight line and its position at any time is given by:

$$x = 3 \cos 2t + 4 \sin 2t$$

(i) Prove that the motion is simple harmonic.

2

(ii) Calculate the particle's greatest speed.

2(c) Water at a temperature of 24°C is placed in a freezer maintained at a temperature of -12°C . After time t minutes the rate of change of temperature T of the water is given by the formula:

$$\frac{dT}{dt} = -k(T + 12)$$

where t is the time in minutes and k is a positive constant.(i) Show that $T = Ae^{-kt} - 12$ is a solution of this equation, where A is a constant.**1**(ii) Find the value of A .**1**(iii) After 15 minutes the temperature of the water falls to 9°C . Find to the nearest minute the time taken for the water to start freezing. (Freezing point of water is 0°C).**3**

(d) Each rectangular table in a hall has nine seats, five facing the front and four facing the back. In how many ways can 9 people be seated at a table if:

(i) Alex and Bella must sit on the same side.

2

(ii) Alex and Bella must sit on opposite sides

2**End of paper**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Frensham 2014**HSC Mathematics Extension 1 Trial HSC Examination****Worked solutions and marking guidelines**

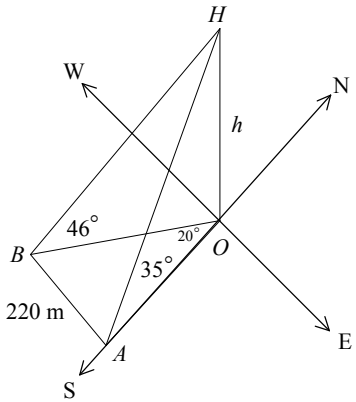
Section I		
	Solution	Criteria
1	Domain: $-1 \leq \frac{x}{2} \leq 1$ or $-2 \leq x \leq 2$. Range: $\frac{1}{2} \times 0 \leq y \leq \frac{1}{2} \times \pi$ or $0 \leq y \leq \frac{\pi}{2}$	1 Mark: C
2	$P(x) = x^3 + ax + 1$ $P(-2) = (-2)^3 + a \times -2 + 1 = 5$ $-2a = 12$ $a = -6$	1 Mark: A
3	$\int \frac{x}{(2-x^2)^3} dx = -\frac{1}{2} \int \frac{1}{u^3} du$ $u = 2 - x^2 \qquad \qquad \qquad = -\frac{1}{2} \times -\frac{1}{2} u^{-2} + C$ $\frac{du}{dx} = -2x$ $-\frac{1}{2} du = x dx \qquad \qquad \qquad = \frac{1}{4(2-x^2)^2} + C$	1 Mark: B
4	$\int_0^1 \frac{1}{x^2+1} dx = [\tan^{-1} x]_0^1$ $= \frac{\pi}{4} - 0$ $= \frac{\pi}{4}$	1 Mark: A
5	Number of arrangements = $\frac{11!}{2! \times 2!}$ (2 I's and 2 B's) $= 9\,979\,200$	1 Mark: B
6	For $y = 2x$ then $m_1 = 2$ For $x + y - 5 = 0$ then $m_2 = -1$	1 Mark: D

	$\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right = \left \frac{2 - (-1)}{1 + 2 \times -1} \right $ $= 3$ $\theta = 71.56505118 \dots$ $\approx 72^\circ$	
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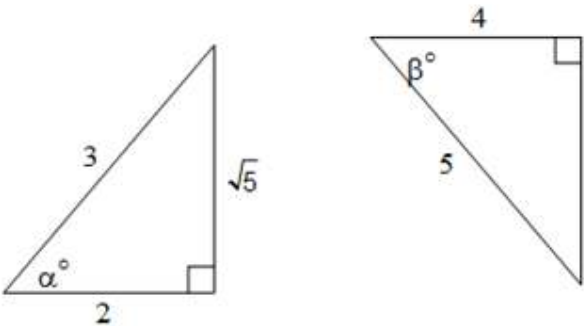
7		1 Mark: A
8	$A(-1,2)$ and $B(3,5)$ with $3:-1$ $x = \frac{mx_2 + nx_1}{m+n} \qquad y = \frac{my_2 + ny_1}{m+n}$ $= \frac{3 \times 3 + -1 \times -1}{3 + -1} \qquad = \frac{3 \times 5 + -1 \times 2}{3 + -1}$ $= 5 \qquad = 6.5$ Point is $(5,6.5)$	1 Mark: D
9	$(x-4)^2 \times \frac{2x-5}{x-4} \geq x \times (x-4)^2 \quad (x \neq 4)$ $(x-4)(2x-5) - x(x-4)^2 \geq 0$ $(x-4)[(2x-5) - x(x-4)] \geq 0$ $(x-4)(-x^2 + 6x - 5) \geq 0$ $(x-4)(x-5)(1-x) \geq 0$ Critical points are 1,4 and 5 or use a sketch and where the	1 Mark: D

	polynomial is above the x-axis. Test values in each region $x \leq 1$ and $4 < x \leq 5$ Note Alternate method: since $x \neq 4$ answer must be B or D then test $x = 0$ in original inequality which gives $\frac{5}{4} > 0$ which is true so $x = 0$ must be included in the solution $\therefore$ D	
10	$v = 2x + 5$ $v^2 = 4x^2 + 20x + 25$ $\frac{1}{2}v^2 = 2x^2 + 10x + \frac{25}{2}$ $a = \frac{d}{dx} \left(2x^2 + 10x + \frac{25}{2} \right)$ $= 4x + 10$ When $x = 1$ then $a = 14$	1 Mark: C

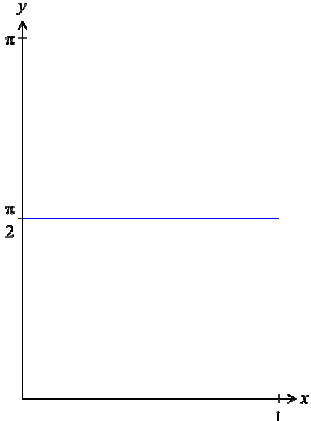
11(c)	$\begin{aligned} \text{LHS} &= \frac{\sin \theta}{\sin \theta + \cos \theta} + \frac{\sin \theta}{\cos \theta - \sin \theta} \\ &= \frac{\sin \theta (\cos \theta - \sin \theta) + \sin \theta (\cos \theta + \sin \theta)}{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)} \\ &= \frac{2 \sin \theta \cos \theta}{\cos^2 \theta - \sin^2 \theta} \\ &= \frac{\sin 2\theta}{\cos 2\theta} \\ &= \tan 2\theta = \text{RHS} \end{aligned}$	2 Marks: Correct answer. 1 Marks: Uses a relevant trigonometric identity
11(d)	Number of ways = ${}^{10}C_3 \times {}^{12}C_2$ $= 120 \times 66$ $= 7920$ Class can be selected in 7920 ways.	2 Marks: Correct answer. 1 Marks: Shows some understanding.

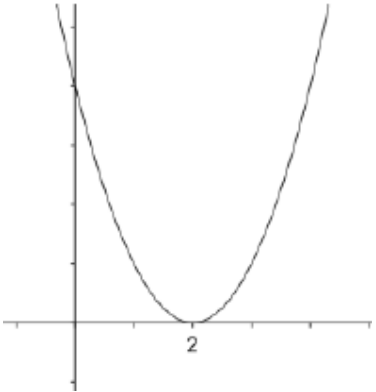
11(e) (i)	 In $\triangle HOA$ $\tan 35^\circ = \frac{h}{OA}$ $OA = \frac{h}{\tan 35^\circ}$ In $\triangle HOB$ $\tan 46^\circ = \frac{h}{OB}$ $OB = \frac{h}{\tan 46^\circ}$	2 Marks: Correct answer. 1 Mark: One correct expression or shows some understanding of the problem.
11(e) (ii)	$AB^2 = OA^2 + OB^2 - 2 \times OA \times OB \times \cos 20^\circ$ $220^2 = \left(\frac{h}{\tan 35^\circ} \right)^2 + \left(\frac{h}{\tan 46^\circ} \right)^2 - 2 \times \frac{h}{\tan 35^\circ} \times \frac{h}{\tan 46^\circ} \times \cos 20^\circ$ $= h^2 \left(\frac{1}{\tan^2 35^\circ} + \frac{1}{\tan^2 46^\circ} - 2 \times \frac{\cos 20^\circ}{\tan 35^\circ \times \tan 46^\circ} \right)$ $h^2 = 220^2 \div \left(\frac{1}{\tan^2 35^\circ} + \frac{1}{\tan^2 46^\circ} - 2 \times \frac{\cos 20^\circ}{\tan 35^\circ \times \tan 46^\circ} \right)$ $= 127296.7453\dots$ $h = 356.7866944\dots \approx 357 \text{ m}$	2 Marks: Correct answer. 1 Mark: Uses the cosine rule with at least one correct value.
11(f)	Factors of 5 are $\{\pm 1, \pm 5\}$ $P(1) = 1^3 + 3 \times 1^2 - 9 \times 1 + 5 = 0$ Therefore $(x-1)$ is a factor of $x^3 + 3x^2 - 9x + 5$ $\begin{array}{r} x^2 + 4x - 5 \\ x-1 \overline{) x^3 + 3x^2 - 9x + 5} \\ \underline{x^3 - x^2} \\ 4x^2 - 9x \\ \underline{4x^2 - 4x} \\ -5x + 5 \\ \underline{-5x + 5} \\ 0 \end{array}$ $P(x) = x^3 + 3x^2 - 9x + 5 = (x-1)(x^2 + 4x - 5)$ $= (x-1)(x-1)(x+5) = (x-1)^2(x+5)$	2 Marks: Correct answer. 1 Mark: Finds one factor or shows some understanding.

12(a) (i)	To find the gradient of the tangent $y = \frac{1}{4a}x^2 \text{ and } \frac{dy}{dx} = \frac{1}{2a}x$ At $P(2ap, ap^2)$ $\frac{dy}{dx} = \frac{1}{2a} \times 2ap = p$ Equation of the tangent at $P(2ap, ap^2)$ $y - y_1 = m(x - x_1)$ $y - ap^2 = p(x - 2ap)$ $y = px - ap^2$ x-intercept ($y = 0$) then $x = ap$. Hence $A(ap, 0)$ y-intercept ($x = 0$) then $y = -ap^2$. Hence $B(0, -ap^2)$ Midpoint of A and B. $M\left(\frac{ap+0}{2}, \frac{0+(-ap^2)}{2}\right) = M\left(\frac{ap}{2}, -\frac{ap^2}{2}\right)$	2 Marks: Correct answer. 1 Mark: Finds the gradient of the tangent or the coordinates of A and B.
12(a) (ii)	To find the locus of M eliminate p from coordinates of M Now $x = \frac{ap}{2}$ (1) and $y = -\frac{ap^2}{2}$ (2) From (1) $p = \frac{2x}{a}$ and sub into eqn (2) $y = \frac{-a\left(\frac{2x}{a}\right)^2}{2} = \frac{-a}{2} \times \frac{4x^2}{a^2} = -\frac{2x^2}{a}$ or $x^2 = -\frac{1}{2}ay$ (parabola)	1 Mark: Correct answer.
12(a) (iii)	$x^2 = -\frac{1}{2}ay = 4 \times \left(-\frac{1}{8}a\right) \times y$ Focus is $\left(0, -\frac{1}{8}a\right)$ and equation of the directrix $y = \frac{1}{8}a$	1 Mark: Correct answer.
12(b)	Step 1: To prove the statement true for $n = 1$ LHS = 1 RHS = $2^1 - 1 = 1$ Result is true for $n = 1$ Step 2: Assume the result true for $n = k$ $1 + 2 + 4 + \dots + 2^{k-1} = 2^k - 1$ Step 3: To prove the result is true for $n = k + 1$ i.e. prove $1 + 2 + 4 + \dots + 2^{k-1} + 2^k = 2^{k+1} - 1$	3 Marks: Correct answer. 2 Marks: Proves the result true for $n = 1$ and attempts to use the result of $n = k$ to prove the result for $n = k + 1$.

	$\begin{aligned} \text{LHS} &= 1 + 2 + 4 + \dots + 2^{k-1} + 2^k \\ &= 2^k - 1 + 2^k \\ &= 2 \times 2^k - 1 && \text{using assumption} \\ &= 2^{k+1} - 1 \\ &= \text{RHS} \end{aligned}$ Result is true for $n = k + 1$ if true for $n = k$ Step 3: Proven true for $n = 1$, assuming true for $n = k$ proven true for $n = k + 1$, so true for $n = 1 + 1 = 2$, $1 + 2 = 3$, and Result true by principle of mathematical induction for all positive integers n.	1 Mark: Proves the result true for $n = 1$.
12(c)	$\sin \left[\cos^{-1} \frac{2}{3} + \tan^{-1} \left(\frac{-3}{4} \right) \right] = \sin \left[\cos^{-1} \frac{2}{3} - \tan^{-1} \frac{3}{4} \right]$ Let $\alpha = \cos^{-1} \frac{2}{3}$ and $\beta = \tan^{-1} \frac{3}{4}$  $\begin{aligned} \sin \left[\cos^{-1} \frac{2}{3} - \tan^{-1} \frac{3}{4} \right] &= \sin \alpha \cos \beta - \cos \alpha \sin \beta \\ &= \frac{\sqrt{5}}{3} \times \frac{4}{5} - \frac{2}{3} \times \frac{3}{5} \\ &= \frac{4\sqrt{5}}{15} \end{aligned}$	2 Marks: Correct answer. 1 Mark: Sets up the two triangles or shows some understanding of the problem.
12(d)	Let $\sin \theta + \cos \theta = R \sin(\theta + \alpha)$ $= R \sin \theta \cos \alpha + R \cos \theta \sin \alpha$ $\therefore R \cos \alpha = 1$ and $R \sin \alpha = 1$ $R^2 (\cos^2 \alpha + \sin^2 \alpha) = 2 \quad \text{and} \quad \tan \alpha = 1 \quad \text{or} \quad \alpha = \frac{\pi}{4}$ $R = \sqrt{2}$	3 Marks: Correct answer. 2 Marks: Finds two angles or makes significant progress towards the solution. 1 Mark: Sets up the sum of two

	$\sin \theta + \cos \theta = \sqrt{2} \sin\left(\theta + \frac{\pi}{4}\right) = 1$ $\sin\left(\theta + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ $\theta + \frac{\pi}{4} = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{9\pi}{4}, \dots$ $\theta = 0, \frac{\pi}{2}, 2\pi$	angles or shows some understanding of the problem.
12(e) (i)	$f(x) = \sin^{-1} x + \cos^{-1} x$ $f'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}} = 0$	1 Mark: Correct answer.

12(e) (ii)	Since $f'(x) = 0$, $f(x)$ is a constant (gradient of tangent is 0) Let $x = 0$ then $f(0) = \sin^{-1} 0 + \cos^{-1} 0 = 0 + \frac{\pi}{2} = \frac{\pi}{2}$ Therefore $f(x) = \frac{\pi}{2}$ for $0 \leq x \leq 1$ 	2 Marks: Correct answer. 1 Mark: Recognises that the graph is a horizontal line or shows some understanding of the problem.
13(a) (i)	$f(x) = xe^x - 1$ $f(0) = 0 \times e^0 - 1 = -1 < 0$ $f(1) = 1 \times e^1 - 1 = e - 1 > 0$ Since $f(0)$ and $f(1)$ have opposite signs and $f(x)$ is a continuous function Therefore the root lies between $x = 0$ and $x = 1$.	1 Mark: Correct answer.
13(a) (ii)	$f(x) = xe^x - 1$ $f'(x) = xe^x + e^x = e^x(x+1)$ $f(0.5) = 0.5e^{0.5} - 1$ $f'(0.5) = e^{0.5}(0.5+1) = 1.5e^{0.5}$ $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ $= 0.5 - \left(\frac{0.5e^{0.5} - 1}{1.5e^{0.5}} \right) = 0.5710204398... \approx 0.57$	2 Marks: Correct answer. 1 Mark: Finds $f(0.5)$, $f'(0.5)$ or shows some understanding of Newton's method.
13(b) (i)	Horizontal Motion $\ddot{x} = 0$ $\dot{x} = c_1$ (when $t = 0, \dot{x} = v \cos 40^\circ$) $\dot{x} = v \cos 40^\circ$ $x = v \cos 40^\circ t + c_2$ (when $t = 0, x = 0$) $x = v \cos 40^\circ t$ (1) Vertical Motion $\ddot{y} = -10$ $\dot{y} = -10t + c_1$ (when $t = 0, \dot{y} = v \sin 40^\circ$) $\dot{y} = -10t + v \sin 40^\circ$ $y = -5t^2 + v \sin 40^\circ t + c_2$ (when $t = 0, y = 0$) $y = -5t^2 + v \sin 40^\circ t$ (2)	3 Marks: Correct answer. 2 Marks: Derives either the horizontal or vertical equations of motion. 1 Mark: States the expressions.

13(b) (ii)	From eqn (1) $t = \frac{x}{v \cos 40^\circ}$ sub into eqn (2) $y = -5 \left(\frac{x}{v \cos 40^\circ} \right)^2 + v \sin 40^\circ \left(\frac{x}{v \cos 40^\circ} \right)$ $= -\frac{5x^2}{v^2} \sec^2 40^\circ + x \tan 40^\circ$	2 Marks: Correct answer. 1 Mark: Eliminates t or shows some understanding.
13(b) (iii)	To find v for $x = 20$ and $y = 6$ $6 = -\frac{5 \times 20^2}{v^2} \sec^2 40^\circ + 20 \times \tan 40^\circ$ $v^2 = \frac{5 \times 20^2 \times \sec^2 40^\circ}{20 \tan 40^\circ - 6}$ $v = 17.77917137 \dots$ $\approx 17.8 \text{ ms}^{-1}$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.
13(c) (i)	 Will have an inverse if strictly increasing or strictly decreasing only. Largest domain, containing $x = 0$ where this occurs is $x \leq 2$	1 Mark: Correct answer.
13(c) (ii)	Domain of $y = f^{-1}(x)$ is the range of $y = f(x)$. Range of $y = f(x)$ is $y \geq 0$. $\therefore$ domain of $y = f^{-1}(x)$ is $x \geq 0$.	1 Mark: Correct answer.
13(c) (iii)	Interchanging x and y, the inverse is $x = (y - 2)^2$ $y - 2 = \pm \sqrt{x}$ $y = 2 \pm \sqrt{x}$ But as $x \leq 2$ for the inverse to exist, $y = 2 - \sqrt{x}$.	1 Mark: Correct answer.
13(c) (iv)	$y = f(x)$ and $y = f^{-1}(x)$ intersect on the line $y = x$. $\therefore y = (x - 2)^2$ and $y = x$ can be solved simultaneously to give the points of intersection for $y = f(x)$ and $y = f^{-1}(x)$. They meet when $x = (x - 2)^2$ i.e. when $x = x^2 - 4x + 4$	2 Marks: correct explanation for why $x = (x - 2)^2$ gives the point of intersection; and correctly solves equation and

	$x^2 - 5x + 4 = 0$ $(x-4)(x-1) = 0$ $x=1$ or 4 But as $x \leq 2$ for the inverse to exist, $y = f(x)$ and its inverse meet when $x=1$.	explains why one solution only. 1 Mark: one of above
14(a)	$\int \cos^2 2x dx = \int \frac{1}{2}(1 + \cos 4x) dx$ $= \frac{1}{2} \left[x + \frac{1}{4} \sin 4x \right] + c$ $= \frac{x}{2} + \frac{1}{8} \sin 4x + c$	2 Marks: Correct answer. 1 Mark: Uses double angle formula.

14(b) (i)	Simple harmonic motion occurs when $\ddot{x} = -n^2x$ Now $x = 3 \cos 2t + 4 \sin 2t$ $\dot{x} = -3 \times 2 \sin 2t + 4 \times 2 \cos 2t$ $\ddot{x} = -3 \times 2^2 \cos 2t - 4 \times 2^2 \sin 2t$ $= -2^2(3 \cos 2t + 4 \sin 2t)$ $\ddot{x} = -2^2x$	2 Marks: Correct answer. 1 Mark: Recognises the condition for SHM.
14(b) (ii)	Maximum speed at $\ddot{x} = 0$ or $x = 0$ (centre of motion) $x = 3 \cos 2t + 4 \sin 2t = 0$ $4 \sin 2t = -3 \cos 2t$ $\tan 2t = -\frac{3}{4}$ $2t = \tan^{-1}(-0.75) + n\pi, \text{ where } n \text{ is an integer}$ $2t = -0.6435011088... + 0, \pi, 2\pi$ Smallest positive value of t for maximum speed $t = \frac{1}{2}(-0.6435011088... + \pi) = 1.249045772...$ $\dot{x} = -3 \times 2 \sin(2 \times 1.24...) + 4 \times 2 \cos(2 \times 1.24...) = -10$ Maximum speed is 10 Alternatively using the auxillary angle method i.e. $v = -6 \sin 2t + 8 \cos 2t$ i.e. $v = 8 \cos 2t - 6 \sin 2t$ now writing this in the form $v = R \cos(2t + \alpha)$ $R = \sqrt{(-6)^2 + (8)^2} = 10$ $\alpha = \tan^{-1}\left(\frac{6}{8}\right)$ $v = 10 \cos(2t + \tan^{-1} 0.75) \text{ which has a maximum value of } 10.$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.
14(c) (i)	$T = Ae^{-kt} - 12 \quad \text{or} \quad Ae^{-kt} = T + 12$ $\frac{dT}{dt} = -kAe^{-kt}$ $= -k(T + 12)$	1 Mark: Correct answer.
14(c) (ii)	Initially $t = 0$ and $T = 24$, $T = Ae^{-kt} - 12$ $24 = Ae^{-k \times 0} - 12$ $A = 36$	1 Mark: Correct answer.
14(c) (iii)	Also $t = 15$ and $T = 9$ $9 = 36e^{-k \times 15} - 12$ $e^{-15k} = \frac{21}{36} = \frac{7}{12}$	3 Marks: Correct answer. 2 Marks: Determines the value of e^{-kt} or makes significant

	$-15k = \log_e \frac{7}{12}$ $k = -\frac{1}{15} \log_e \frac{7}{12}$ $= 0.03593310005\dots$ We need to find t when $T = 0$	progress. 1 Mark: Finds the exact value of k or shows some understanding.
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	$0 = 36e^{-kt} - 12$ $e^{-kt} = \frac{12}{36} = \frac{1}{3}$ $-kt = \log_e \frac{1}{3}$ $t = -\frac{1}{k} \log_e \frac{1}{3}$ $= 30.5738243... \approx 31 \text{ minutes}$ It will take about 31 minutes for the water to cool to 0°C	
14(d) (i)	Facing front: Number of ways = $5 \times 4 \times 7!$ Facing back: Number of ways = $4 \times 3 \times 7!$ Total number of ways = $(5 \times 4 + 4 \times 3) \times 7!$ $= 161\,280$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.
14(d) (ii)	Alex facing front and Bella facing back Number of ways = $5 \times 4 \times 7!$ Bella facing front and Alex facing back Number of ways = $5 \times 4 \times 7!$ Total number of ways = $(5 \times 4 \times 7!) \times 2$ $= 201\,600$	2 Marks: Correct answer. 1 Mark: Makes some progress towards the solution.